

Homework I - Sens ERSOY

060200434

1) $\vec{R}$: Resultant Force

$\sum \vec{F}_i$: Sum of individual forces

m : mass

$\vec{a}_c$: acceleration of the center of mass

M_0 = Moment at the reference point

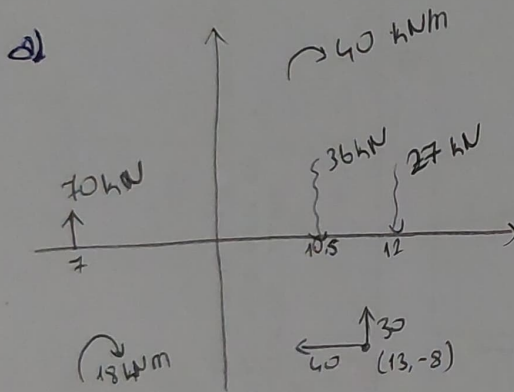
$\sum \vec{M}_i$: Sum of individual moments
(Force couples)

$\sum r_{io} \times F_i$: Moment of the individual forces to an arbitrary point

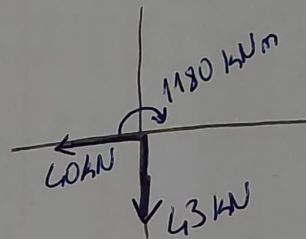
I_0 : Moment of Inertia

α : Angular acceleration

2)



$$390 - 378 - 324 - 320 - 490 - 40 - 18 = -1180 \text{ kNm}$$

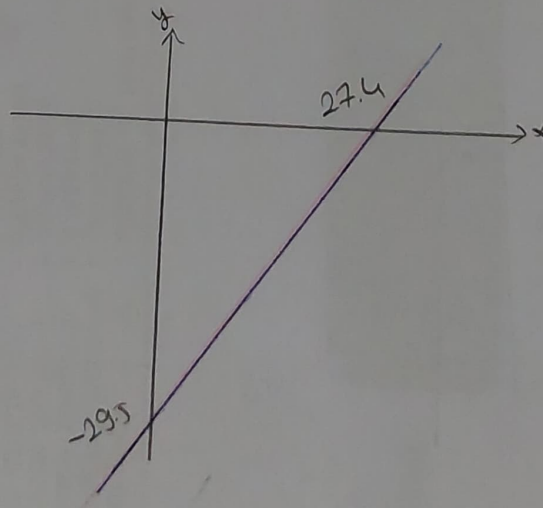


b)

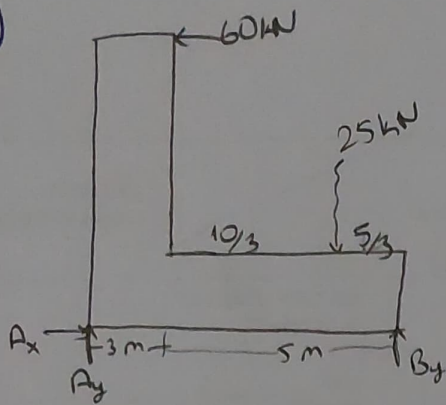
$$40y - 43x = -1180$$

$$y = -29.5$$

$$x = 27.4$$



3)



$$A_x = 60 \text{ kN}$$

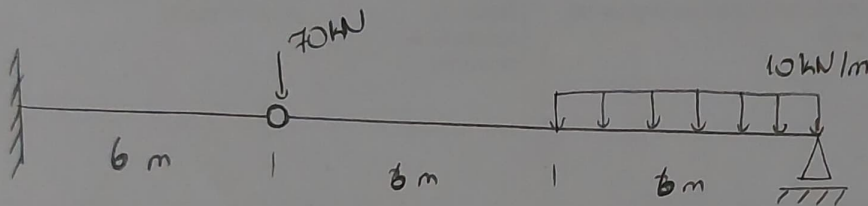
$$A_y + B_y = 25 \text{ kN}$$

$$8 B_y + 60 \cdot 8 - 25 \cdot \frac{19}{3} = 0$$

$$B_y = -40,2$$

$$A_y = 65,2$$

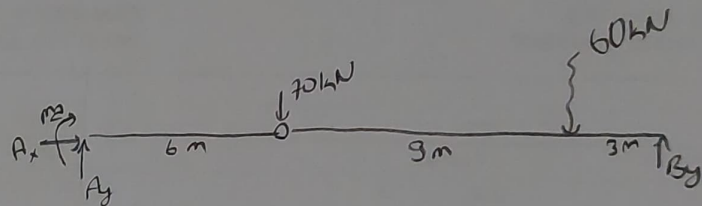
4)



$$12 B_y - 60 \cdot 9 = 0$$

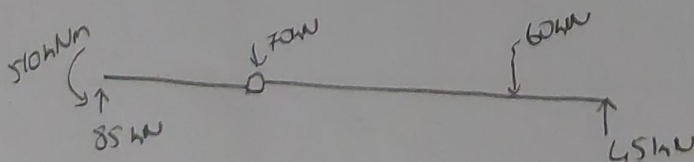
$$B_y = 45 \text{ kN}$$

$$A_y = 85 \quad A_x = 0$$



$$-70 \cdot 6 - 60 \cdot 15 + 45 \cdot 18 - m_b = 0$$

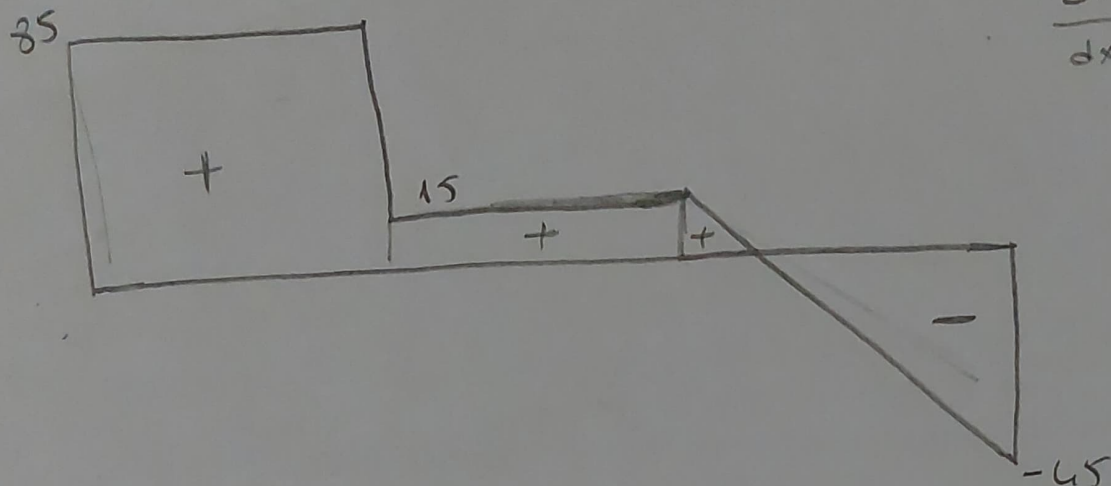
$$m_b = -510 \text{ kNm}$$



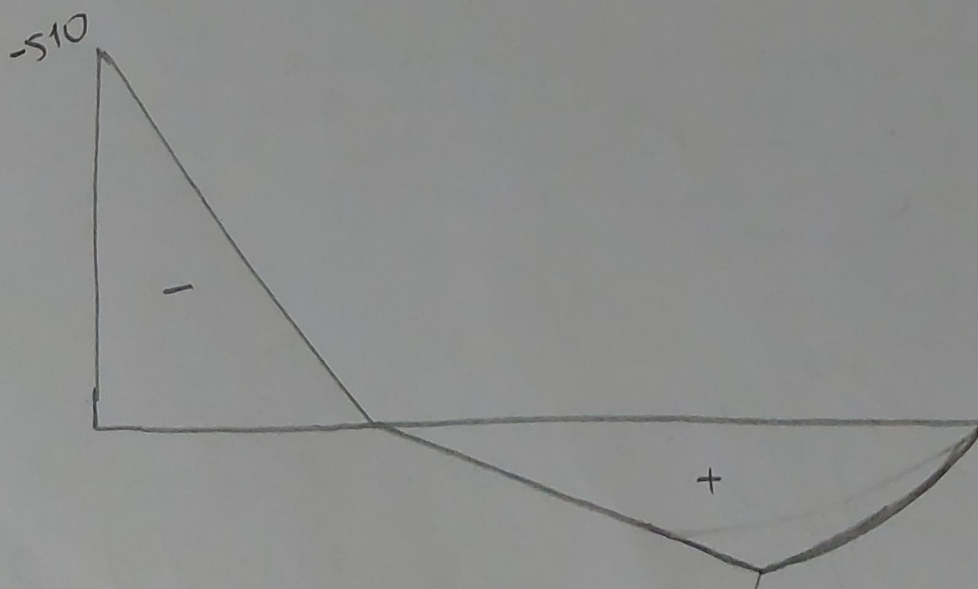
T

$$\frac{dT_y}{dx} = -q_y$$

$$\frac{dM_x}{dx} = T_y$$

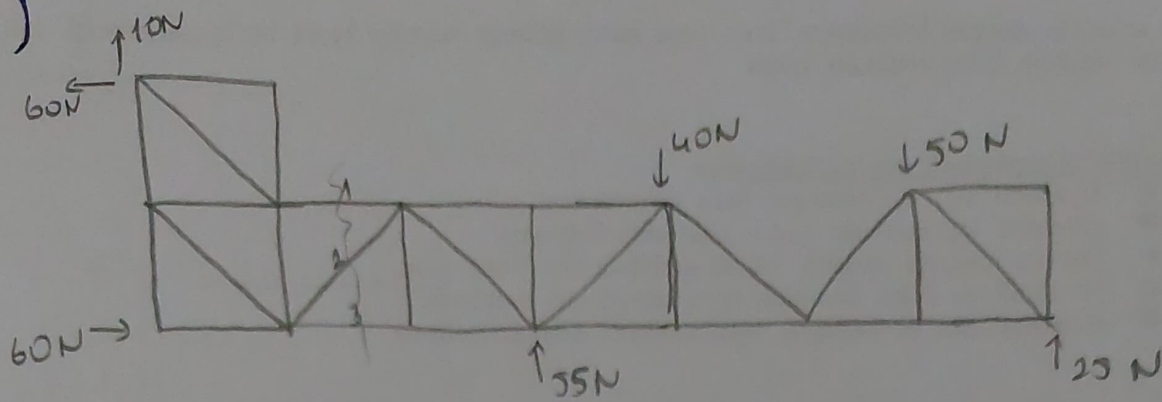


M



90, The zero of
the T diagram

5)

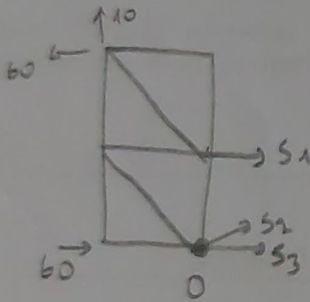


$$60 \cdot 4 + 6 B_y - 40 \cdot 8 - 50 \cdot 12 + 25 \cdot 14 = 0$$

$$B_y = 55 \text{ N}$$

$$4 C_y - 100 = 0$$

$$C_y = 25 \text{ N}$$



$$M_O = -S_1 \cdot 2 + 60 \cdot 4 - 10 \cdot 2 = 0$$

$$S_1 = 110 \text{ (tension)}$$

$$S_2 \sin \alpha = \frac{S_2}{\sqrt{2}} = -10$$

$$S_2 = -10\sqrt{2} \text{ (compression)}$$

$$S_3 + 110 + 60 - 60 - 10 = 0$$

$$S_3 = -100 \text{ (compression)}$$